

Poster: High-Fidelity and Contextual User Activity Memory Forensics

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Abstract

Retrieving application user activities from a memory dump has traditionally been a labor-intensive process, due to both the sheer volume of memory data and the complexity of opaque data structures. We introduce RAM-Weaver, a system that enables contextual querying of user activities from application memory dump files. RAM-Weaver distills essential information from raw dumps and leverages large language models (LLMs) to navigate opaque data structures and support interactive queries. This distillation process can reduce the data volume by up to 99.9% and improve the signal-to-noise ratio by nearly 37 dB, making it feasible to run on smaller, offline models. When combined with full-scale, cloud-hosted models, RAM-Weaver can retrieve user activity data with exceptionally high accuracy.

CCS Concepts

• Applied computing → Computer forensics → Evidence collection, storage and analysis.

Keywords

Memory Forensics, Preprocessing, Data Reconstruction, Large Language Models, Contextual Analysis, Noise Reduction

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1 Introduction

The process memory of an application program is a critical source of evidence in digital forensics, often containing a wealth

of volatile data that never gets written to disk. This "digital crime scene" can hold transient artifacts like real-time chat messages, unsaved document contents, and credentials, making its analysis essential for reconstructing user activities [1].

However, distilling evidentiary value from this repository of volatile data is fundamentally impeded by its fragmented, unstructured, and ephemeral nature. Traditional methods, like running strings [2] or grep [3], often result in a deluge of irrelevant noise, burying valuable artifacts in thousands of fragmented and context-free character sequences. More sophisticated techniques involve manually reverse-engineering application-specific data structures. While precise, this approach is extremely labor-intensive, brittle to software updates, and fails to scale across the vast ecosystem of modern applications. Although machine learning has been applied to memory forensics, it has primarily focused on classification tasks such as malware detection and anomaly identification [4, 5]. Its potential for reconstructing high-fidelity, contextual user text from the scattered data layouts of complex applications remains largely unexplored. The core challenge has shifted from merely finding strings to comprehending and reassembling fragmented information into a coherent narrative.

To address these challenges, we introduce RAM-Weaver, a novel framework that reimagines memory artifact reconstruction as a natural language processing and generation task. Instead of relying on rigid structures, RAM-Weaver leverages the powerful pattern recognition and reasoning capabilities of Large Language Models (LLMs) built upon the Transformer architecture [6].

Our primary contributions are as follows: (1) we propose **RAM-Weaver**, a framework that pioneers the use of LLMs for high-fidelity text reconstruction in memory forensics, shifting from data carving to contextual understanding; (2) we develop **Adaptive Memory Carver (AMC)** for targeted, application-aware data extraction to overcome the prohibitive size of memory dumps for LLM analysis; (3) we provide a detailed case study on the social super app LINE Messenger [7] popular in Asian countries, demonstrating RAM-Weaver's ability to accurately **restore conversations** from memory dumps where traditional methods fail; (4) we show our approach enables analysts to perform **contextual forensic queries**, a capability unattainable with previous methods.

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2 The RAM-Weaver Framework

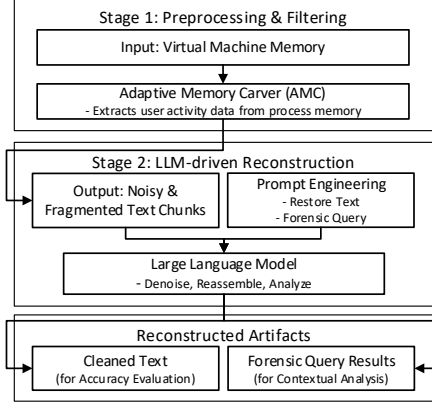


Figure 1: Overview of the RAM-Weaver workflow for artifact reconstruction.

Our framework, RAM-Weaver (Figure 1), is a two-stage pipeline designed to overcome the primary obstacle in applying LLMs to memory forensics: the infeasibly large and noisy memory dumps, which far exceed LLM context windows.

Stage 1: Preprocessing & Filtering. Executed by our Adaptive Memory Carver (AMC), this stage intelligently reduces data volume while preserving forensic artifacts, outputting condensed text chunks. AMC performs two key functions: (1) Adaptive Memory Extraction (2) Targeted Artifact Filtering.

Stage 2: LLM-driven Reconstruction. Extracted fragments are fed into an LLM. Guided by Prompt Engineering, the LLM handles the following forensic tasks: (1) High-Fidelity Restoration to denoise and reassemble the original text, and (2) Contextual Forensic Querying that analyzes content and provides answers interactively in response to investigation questions.

2.1 Adaptive Memory Carver (AMC)

The AMC extracts text chunks from a memory dump through a flexible, two-step process: adaptive memory extraction followed by targeted artifact filtering.

Adaptive Memory Extraction (AME). The AMC first selects an extraction strategy tailored to the target application's memory allocation. It supports two primary modes: a Heap Mode that parses the Process Environment Block (PEB) to dump heap-containing VAD (Virtual Address Descriptor) regions precisely, and a broader PrivateMemory Mode that traverses the VAD tree to dump all regions marked as PrivateMemory, while intelligently excluding memory mapped from executable files (VadImageMap) or devices.

Targeted Artifact Filtering. The extracted binary data is then passed to a filtering layer where targeted filter rules are applied. The current filter rules focus on ASCII/UTF-8 and UTF-16LE encoded contents: (1) Regex-based Filtering eliminates system-related noise (e.g., file paths, GUIDs) from unstructured text. (2) JSON-like Pattern Filtering identifies key-value patterns

(e.g., "text":"...") in semi-structured data, crucially surrounding context to preserve valuable structural information for the LLM.

3 Experimental Setup

We evaluated RAM-Weaver's effectiveness on a real-world application, LINE Messenger (v9.9.0.3633), running in a Windows 10 guest virtual machine. For this program process, the AMC automatically selected its PrivateMemory Mode and JSON-like filtering strategy. We designed three distinct scenarios to test our framework's capabilities: **(S1) AMC Efficacy**, which compared our full pipeline against a naïve 'strings' baseline and a hybrid approach to demonstrate its necessity, using the Amazon Reviews dataset [8]. For both S1 and S2, we developed a semi-automated script that posted each review as a separate message into a LINE chat. After each message was sent, we captured the system memory dump for analysis; **(S2) Restoration Accuracy**, which measured reconstruction fidelity on single messages, also using the Amazon Reviews dataset [8]; and **(S3) Forensic Querying**, which evaluated contextual analysis on a multi-turn conversation from the Topical-Chat dataset [9]. Similar to S2, a script was used to simulate a conversation by sending messages from the dataset between two user accounts in LINE.

We employed a corresponding set of evaluation metrics. For **Preprocessing Efficacy (S1)**, we measured Input Size (raw size of text fed to the LLM), Signal-to-Noise Ratio (SNR), Feasibility (LLM processability) and Character Error Rate (CER) both before (Initial) and after (Final) the LLM's restoration. For **Restoration Accuracy (S2)**, we assessed performance using Exact Match Rate (EMR, percentage of reconstructed texts that were identical to the original message) and CER (Levenshtein distance at the character level, providing a more granular measure of similarity for near-perfect reconstructions). Finally, for **Forensic Querying (S3)**, we qualitatively illustrate a response's Correctness (accuracy of retrieved information) and Completeness (if all requested information was returned).

4 Results and Analysis

4.1 AMC Efficacy: An Essential Prerequisite (S1)

Table 1: Impact of the AMC on Preprocessing and Final Restoration

Method	Input Data Size ↓	Signal-to-Noise Ratio (SNR) ↑	LLM Feasibility	Initial CER (Pre-LLM) ↓	Final CER (Post-LLM) ↓
Baseline 1 (Process Memory + Naïve strings)	~34.10 MB	-47.33 dB	Infeasible (Context Limit)	57416.48	N/A
Baseline 2 (AMC's AME + Strings)	~5.58 MB	-39.41 dB	Infeasible (Context Limit)	9537.20	N/A
RAM-Weaver (Full AMC pipeline)	~15.80 KB	-10.53 dB	Feasible	18.61	0.20

The AMC is fundamental to making LLM analysis of memory dumps viable. To demonstrate its necessity, we compared our full AMC pipeline against two baselines. As shown in Table 1, both baselines failed: a naïve `strings` approach produced 34.10 MB of text, and a hybrid approach still yielded 5.58 MB. Both were too large for the LLMs, rendering them "Infeasible."

In stark contrast, our full AMC pipeline was the only method to produce a "Feasible" input, reducing the data size by over 99.9% to just 15.80 KB. This was achieved by improving the SNR by nearly 37 dB. The direct impact on quality is evident in the Initial CER, which the AMC lowered from an unusable >57,000 to a manageable 18.61. This clean input enabled the LLM to achieve a final, accurate reconstruction (Final CER of 0.20), confirming that the AMC transforms an otherwise impossible task into a solvable one.

4.2 Single Message Restoration Accuracy (S2)

With the AMC's high-quality output, we evaluated various LLMs on the reconstruction of single messages. As shown in Table 2, performance varied dramatically. OpenAI's GPT-3 achieved a flawless 100% EMR, setting a benchmark for perfect restoration. In contrast, Google's Gemini 2.5 Pro achieved only 60% EMR, primarily due to struggles with quotation mark fidelity, which often resulted in the swapping of single and double quotes.

Table 2: Single Message Restoration Accuracy on LINE

Model Family	Model	Exact Match Rate (Avg. EMR) ↑	Character Error Rate (Avg. CER) ↓
Commercial	OpenAI GPT-o3	100%	0.00
	Google Gemini 2.5 Pro	60%	0.20
Open-Source	Google Gemma 3 (27b)	40%	4.32
	Google Gemma 3 (12b)	0%	7.83
	Google Gemma 3 (4b)	30%	2.11

The open-source Gemma 3 models revealed a different core weakness: a persistent difficulty in deduplicating text fragments common in memory dumps, which degraded their EMR and CER scores. Interestingly, performance did not scale with model size; the small 4b model (30% EMR) surprisingly outperformed the much larger 12b model (0% EMR). This suggests smaller models may be less prone to over-interpreting noisy, fragmented inputs on this specific task.

4.3 Contextual Forensic Querying (S3)

RAM-Weaver's ability to perform contextual analysis on conversation history showcases its power beyond mere text restoration. As illustrated in Figure 2, it elevates the process from data carving to forensic reasoning. The input is a raw stream of fragments with cryptic user IDs and machine-readable timestamps.

When given a simple natural language prompt, the LLM does not just find strings. It interprets intent, parses data structures, converts timestamps, identifies speakers, and synthesizes the results into a coherent, human-readable answer. This ability to directly transform a chaotic data mess into a precise forensic log

that fulfills the investigator's intent represents a paradigm shift in memory forensics.

Input Data from AMC	Forensic Querying
<pre> {"from": "u81c9pf8f6vyhynods8yaqgahw1y0kxp3", "to": "u81f4ippue35w6bf0x9kks2p62thzma8", "type": "0", "id": "568086353582752223", "createdTime": "175143715719", "hasContent": false, "contentType": "0", "contentMetadata": {"e2eeMark": "1"}, "sessionId": "0", "chunks": [], "reactions": [], "text": "Yes and they smile more when they hit the beat", "location": "0", "type": "1", "status": "2", "chatId": "u81f4ippue35w6bf0x9kks2p62thzma8", "readCount": "0", "reqSeqV2": "42", "reqSeq": "0", "contentInfo": {"eventInfo": {"rev": "1795", "errorCode": "0", "urlPreview": "0", "hasUrlPreview": false, "syncToken": ""}, "fromType": "0", "attribute": "0", "flag": "0", "reactionStatus": "0", "decrypted": true}, "text": "查看使用中的会员优惠", "url": "https://line.me/R/mv/settings/linePremium/status?entry=hometa_b_fixed"} </pre>	<pre> > Prompt: "List all messages after Wed Jul 02 2025 14:15:22 (Taipei time)."</pre>
<pre> {"from": "u81f4ippue35w6bf0x9kks2p62thzma8", "to": "u81c9pf8f6vyhynods8yaqgahw1y0kxp3", "type": "0", "id": "568085960609497277", "createdTime": "175143692290", "hasContent": false, "contentType": "0", "contentMetadata": {"e2eeMark": "2"}, "sessionId": "0", "chunks": [], "reactions": [], "text": "Yeah. Ballet dancer go through 4 pairs of shoes a week", "location": "0", "type": "1", "status": "1", "chatId": "u81f4ippue35w6bf0x9kks2p62thzma8", "readCount": "0", "reqSeqV2": "0", "reqSeq": "0", "contentInfo": {"eventInfo": {"rev": "1789", "errorCode": "0", "urlPreview": "0", "hasUrlPreview": false, "syncToken": ""}, "fromType": "0", "attribute": "0", "flag": "0", "reactionStatus": "0", "decrypted": true}, "text": "Yeah. Ballet dancer go through 4 pairs of shoes a week"} </pre>	<pre> [14:15:22] u81f4ippue35w6bf0x9kks2p62thzma8: Yeah. Ballet dancer go through 4 pairs of shoes a week [14:16:39] u61c9pf8f6vyhynods8yaqgahw1y0kxp3: Yes that is a lot of shoes and also a lot of money [14:18:02] u81f4ippue35w6bf0x9kks2p62thzma8: Yeah true. Did you know babies are really good at dancing? [14:20:39] u61c9pf8f6vyhynods8yaqgahw1y0kxp3: Yes and they smile more when they hit the beat [14:20:29] u81f4ippue35w6bf0x9kks2p62thzma8: Yeah they are much smarter than we give them credit for </pre>
<pre> ... more fragmented data ... </pre>	

Figure 2: Example of LLM-driven Forensic Querying.

5 Conclusion

RAM-Weaver enables efficient, contextual querying of user activities from application memory dumps. Its Adaptive Memory Carver (AMC) dramatically reduces data size by over 99.9% and improves the signal-to-noise ratio (SNR) by nearly 37 dB. Experimental results demonstrate that RAM-Weaver achieves perfect restoration accuracy (100% EMR) when using full-scale models such as GPT-o3. While smaller, locally-hosted models like Google Gemma 3 (4b) demonstrate feasibility with a 30% EMR, showcasing the potential for offline analysis. The proposed interactive query approach not only streamlines forensic investigations but also holds promise for broader security applications, including user privacy monitoring.

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